

SIMATS ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Analytical Problem

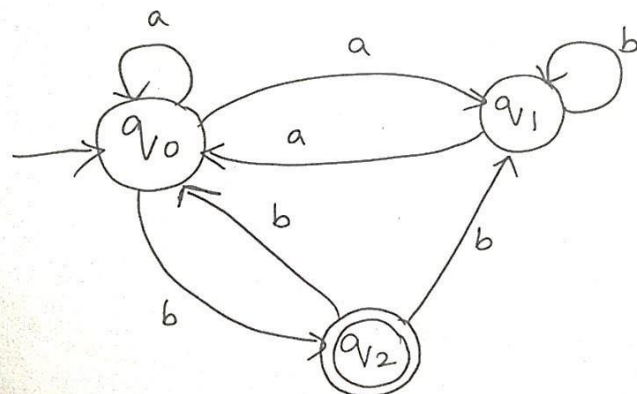
Code of Conduct:

I J.Goutham Kumar Reddy 192311140 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L



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ANS)

The **Subset Construction Algorithm** to convert the Non-Deterministic Finite Automaton (NFA) into an equivalent Deterministic Finite Automaton (DFA). **Step 1: Extract the NFA Transition Table**

First, let's identify the formal components of the NFA from the image:

States (Q): {q_0, q_1, q_2 }

Alphabet (Sigma): {a, b }

Start State (q_0): q_0

Accepting State (F): {q_2 } (indicated by the double circle) the transition table (δ_{NFA}) derived from the drawing:

State	Input a	Input b
$\rightarrow q_0$	{q_0, q_1 }	{q_2 }
q_1	{q_0 }	{q_1 }
* q_2	emptyset	{q_0, q_1 }

Step 2: Apply Subset Construction

To build the DFA, each state will represent a subset of the NFA's states. We start with the initial state and generate new states as we evaluate the transitions for both inputs a and b .

1. Initial State (A):

The DFA start state is the set containing the NFA start state.

Let $A = \{q_0\}$

On input a : $\delta(A, a) = \delta(q_0, a) = \{q_0, q_1\}$. This is a new subset, let's call it **B** . On

input b : $\delta(A, b) = \delta(q_0, b) = \{q_2\}$. This is a new subset, let's call it **C** .

2. Evaluate State B ({q_0, q_1}):

To find the transitions for a subset, we take the union of the transitions of its individual NFA states.

On input a : $\delta(B, a) = \delta(q_0, a) \cup \delta(q_1, a) = \{q_0, q_1\} \cup \{q_0\} = \{q_0, q_1\}$. This is state **B** .

On input b : $\delta(B, b) = \delta(q_0, b) \cup \delta(q_1, b) = \{q_2\} \cup \{q_1\} = \{q_1, q_2\}$. This is a new subset, let's call it **D** .

3. Evaluate State C ({q_2}):

On input a : $\delta(C, a) = \delta(q_2, a) = \text{emptyset}$. This is the empty set (dead state), let's call it **E** . On

input b : $\delta(C, b) = \delta(q_2, b) = \{q_0, q_1\}$. This is state **B**

4. Evaluate State D ({q_1, q_2}):

On input a : $\delta(D, a) = \delta(q_1, a) \cup \delta(q_2, a) = \{q_0\} \cup \text{emptyset} = \{q_0\}$. This is state **A** .

On input b : $\delta(D, b) = \delta(q_1, b) \cup \delta(q_2, b) = \{q_1\} \cup \{q_0, q_1\} = \{q_0, q_1\}$. This is state **B** .

5. Evaluate State E (emptyset - Dead State):

On input a : $\delta(\text{emptyset}, a) = \text{emptyset}$. This is state **E** .

On input b : $\delta(\text{emptyset}, b) = \text{emptyset}$. This is state **E**

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Step 3: Define the Final DFA

The accepting states in our new DFA are any state subsets that contain the original NFA's accepting state (q_2). Therefore, states C and D are accepting states. Here is the finalized DFA transition table:

DFA State (Subset)	Input a	Input b
$\rightarrow A (\{q_0\})$	B	C
$B (\{q_0, q_1\})$	B	D
$* C (\{q_2\})$	E	B
$* D (\{q_1, q_2\})$	A	B
$E (\text{emptyset})$	E	E

2. Let L be the language of words of length at least 2 whose last two letters are different from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

ANS)

The step-by-step construction of the Non-Deterministic Finite Automaton (NFA) and its equivalent Deterministic Finite Automaton (DFA).

Part 1: Constructing the NFA

To design the NFA for the regular expression $(a+b)^*(ab+ba)$, we need the automaton to nondeterministically "guess" when it has reached the second-to-last character of the string.

States (Q): $\{q_0, q_1, q_2, q_f\}$

Alphabet (Sigma): $\{a, b\}$

Start State: q_0

Accepting State (F): $\{q_f\}$

NFA Logic:

q_0 will loop on both a and b to consume the prefix of the string.

If q_0 reads an a, it can non-deterministically transition to q_1 (guessing it's the start of the final ab).

If q_0 reads a b, it can non-deterministically transition to q_2 (guessing it's the start of the final ba).

From q_1 , reading a b goes to the accepting state q_f .

From q_2 , reading an a goes to the accepting state q_f .

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NFA Transition Table:

State	Input: a	Input: b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	emptyset	$\{q_f\}$
q_2	$\{q_f\}$	emptyset
$* q_f$	emptyset	emptyset

Part 2: Constructing the Equivalent DFA

To convert this NFA to a DFA, we use the **Subset Construction Algorithm**. Each state in the new DFA will represent a subset of states from the NFA.

Initial State (A):

Start with the NFA's start state: $A = \{q_0\}$.

On a : $\delta(\{q_0\}, a) = \{q_0, q_1\}$. Call this new state **B**.

On b : $\delta(\{q_0\}, b) = \{q_0, q_2\}$. Call this new state **C**.

Evaluate State B ($\{q_0, q_1\}$):

On a : $\delta(q_0, a) \cup \delta(q_1, a) = \{q_0, q_1\} \cup \text{emptyset} = \{q_0, q_1\}$. This is state **B**.

On b : $\delta(q_0, b) \cup \delta(q_1, b) = \{q_0, q_2\} \cup \{q_f\} = \{q_0, q_2, q_f\}$. Call this new state **D**.

Evaluate State C ($\{q_0, q_2\}$):

On a : $\delta(q_0, a) \cup \delta(q_2, a) = \{q_0, q_1\} \cup \{q_f\} = \{q_0, q_1, q_f\}$. Call this new state **E**.

On b : $\delta(q_0, b) \cup \delta(q_2, b) = \{q_0, q_2\} \cup \text{emptyset} = \{q_0, q_2\}$. This is state **C**.

Evaluate State D ($\{q_0, q_2, q_f\}$):

On a : $\delta(q_0, a) \cup \delta(q_2, a) \cup \delta(q_f, a) = \{q_0, q_1\} \cup \{q_f\} \cup \text{emptyset} = \{q_0, q_1, q_f\}$. This is state **E**.

On b : $\delta(q_0, b) \cup \delta(q_2, b) \cup \delta(q_f, b) = \{q_0, q_2\} \cup \text{emptyset} \cup \text{emptyset} = \{q_0, q_2\}$. This is state **C**.

Evaluate State E ($\{q_0, q_1, q_f\}$):

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On a : $\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_f, a) = \{q_0, q_1\} \cup \emptyset \cup \emptyset = \{q_0, q_1\}$. This is state **B** .

On b : $\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_f, b) = \{q_0, q_2\} \cup \{q_f\} \cup \emptyset = \{q_0, q_2, q_f\}$. This is state **D** .

Identifying Accepting States:

Any subset containing the NFA's accepting state (q_f) becomes an accepting state in the DFA. Therefore, states **D** and **E** are accepting.

Equivalent DFA Transition Table:

DFA State (Subs	State Meaning (En	Input: a	Input: b
$\rightarrow A (\{q_0\})$	Empty or irrelevant	B	C
$B (\{q_0, q_1\})$	Ends in a (or aa)	B	D
$C (\{q_0, q_2\})$	Ends in b (or bb)	E	C
$* D (\{q_0, q_2, q_f\})$	Ends in ab (Accepted)	E	C
$* E (\{q_0, q_1, q_f\})$	Ends in ba (Accepted)	B	D

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

ANS)

The design and formal definition of the Nondeterministic Finite Automaton (NFA) that accepts strings ending in 00 or 11, followed by the trace proving the acceptance of the string 01011.

Formal Definition of the NFA

An NFA is formally defined as a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$:

States (Q): $Q = \{q_0, q_1, q_2, q_f\}$ q_0 : The initial state, which loops to consume the prefix of the string. q_1 : The state reached when the NFA non-deterministically "guesses" a 0 is the first part of a 00 ending. q_2 : The state reached when the NFA non-deterministically "guesses" a 1 is the first part of a 11 ending. q_f : The accepting final state, reached after successfully observing 00 or 11.

Alphabet (Sigma): $\Sigma = \{0, 1\}$

Initial State (q_0): q_0

Accepting States (F): $F = \{q_f\}$

Transition Function (delta): $\delta : Q \times \Sigma \rightarrow 2^Q$

Transition Table (delta)

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State	Input: 0	Input: 1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_f\}$	emptyset
q_2	emptyset	$\{q_f\}$
$* q_f$	emptyset	emptyset

Acceptance of the String "01011"

For an NFA to accept a string, there must exist **at least one** valid path from the initial state to an accepting state that consumes the entire input string. Because it is non-deterministic, multiple paths are explored simultaneously.

Let's trace the execution of the input string 01011 step-by-step to see the set of active states at each point:

Start: The NFA begins in the initial state. Current States: $\{q_0\}$

Read 0:

From q_0 , on input 0, the NFA transitions to $\{q_0, q_1\}$.

Current States: $\{q_0, q_1\}$

Read 1:

From q_0 on 1, it goes to q_0 and q_2 .

From q_1 on 1, there is no valid transition (it dies/returns emptyset) Current

States: $\{q_0, q_2\}$

Read 0:

From q_0 on 0, it goes to q_0 and q_1 .

From q_2 on 0, there is no valid transition (it dies/returns emptyset).

Current States: $\{q_0, q_1\}$

Read 1:

From q_0 on 1, it goes to q_0 and q_2 .

From q_1 on 1, there is no valid transition (it dies/returns emptyset). Current

States: $\{q_0, q_2\}$

Read 1:

From q_0 on 1, it goes to q_0 and q_2 .

From q_2 on 1, it successfully transitions to the accepting state q_f . Final

States: $\{q_0, q_2, q_f\}$

After processing the entire string 01011, the set of active states is $\{q_0, q_2, q_f\}$. Because this resulting set contains the accepting state q_f ($q_f \in F$), the string **is accepted** by the NFA.

The successful accepting path out of the non-deterministic choices is: $q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$

q_f

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0,1\}$. Construct a DFA that represents the same language.

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Although you didn't include an image of the specific NFA this time, the standard NFA for the language of all strings containing the substring "001" over the alphabet $\Sigma = \{0, 1\}$ is well-defined.

I will outline this standard NFA and then use the **Subset Construction Algorithm** to convert it into the equivalent DFA.

Step 1: The Standard NFA for the substring "001"

The NFA works by non-deterministically guessing when the substring "001" begins.

States: $Q = \{q_0, q_1, q_2, q_3\}$

Start State: q_0 **Accepting**

State: q_3

NFA Transition Table:

State	Input: 0	Input: 1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	emptyset
q_2	emptyset	$\{q_3\}$
$* q_3$	$\{q_3\}$	$\{q_3\}$

Step 2: Subset Construction

We create the DFA by generating states that represent subsets of the NFA's states.

1. Initial State (A):

Start with the subset containing the NFA's start state: $A = \{q_0\}$.

On 0: $\delta(\{q_0\}, 0) = \{q_0, q_1\}$. Call this new state **B**

On 1: $\delta(\{q_0\}, 1) = \{q_0\}$. This is state **A**.

2. Evaluate State B ({q_0, q_1}):

On 0: $\delta(q_0, 0) \cup \delta(q_1, 0) = \{q_0, q_1\} \cup \{q_2\} = \{q_0, q_1, q_2\}$. Call this new state **C**.

On 1: $\delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0\} \cup \text{emptyset} = \{q_0\}$. This is state **A**.

3. Evaluate State C ({q_0, q_1, q_2}):

On 0: $\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0) = \{q_0, q_1\} \cup \{q_2\} \cup \text{emptyset} = \{q_0, q_1, q_2\}$. This is state **C**.

On 1: $\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1) = \{q_0\} \cup \text{emptyset} \cup \{q_3\} = \{q_0, q_3\}$. Call this new state **D**.

4. Evaluate State D ({q_0, q_3}):

On 0: $\delta(q_0, 0) \cup \delta(q_3, 0) = \{q_0, q_1\} \cup \{q_3\} = \{q_0, q_1, q_3\}$. Call this new state **E**.

On 1: $\delta(q_0, 1) \cup \delta(q_3, 1) = \{q_0\} \cup \{q_3\} = \{q_0, q_3\}$. This is state **D**.

5. Evaluate State E ({q_0, q_1, q_3}):

On 0: $\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_3, 0) = \{q_0, q_1\} \cup \{q_2\} \cup \{q_3\} = \{q_0, q_1, q_2, q_3\}$. Call this new state **F**.

On 1: $\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_3, 1) = \{q_0\} \cup \text{emptyset} \cup \{q_3\} = \{q_0, q_3\}$. This is state **D**.

6. Evaluate State F ({q_0, q_1, q_2, q_3}):

On 0: $\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0) \cup \delta(q_3, 0) = \{q_0, q_1\} \cup \{q_2\} \cup \text{emptyset} \cup \{q_3\} = \{q_0, q_1, q_2, q_3\}$. This is state **F**.

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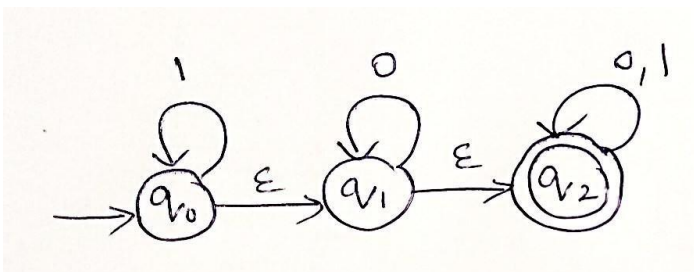
On 1: $\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1) \cup \delta(q_3, 1) = \{q_0\} \cup \emptyset \cup \{q_3\} \cup \{q_3\} = \{q_0, q_3\}$. This is state **D**.

Step 3: Define the Final DFA

Any subset that contains the NFA's accepting state (q_3) becomes an accepting state in the DFA. Therefore, states **D**, **E**, and **F** are all accepting states. **Equivalent DFA Transition Table:**

DFA State (Subset)	State Meaning (Prefix)	Input: 0	Input: 1
$\rightarrow A (\{q_0\})$	Empty or saw '1'	B	A
B ($\{q_0, q_1\}$)	Saw '0'	C	A
C ($\{q_0, q_1, q_2\}$)	Saw '00'	C	D
* D ($\{q_0, q_3\}$)	Saw '001' (Accepted)	E	D
* E ($\{q_0, q_1, q_3\}$)	Saw '001', then '0' (Accepted)	F	D
* F ($\{q_0, q_1, q_2, q_3\}$)	Saw '001', then '00' (Accepted)	F	D

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



ANS)

two main steps: first computing the epsilon -closures, and then using the subset construction algorithm to build the Deterministic Finite Automaton (DFA).

Step 1: Find the epsilon -Closure for Each State

The epsilon -closure of a state is the set of all states that can be reached from it by making zero or more epsilon -transitions (without consuming any input symbols).

epsilon text{-closure}(q_0): From q_0 , we can reach q_1 via ϵ , and from q_1 we can reach q_2 via ϵ . Including the state itself, the closure is:

$$\epsilon \text{ text{-closure}}(q_0) = \{q_0, q_1, q_2\}$$

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$\epsilon\text{-closure}(q_1)$: From q_1 , we can reach q_2 via ϵ . Including the state itself

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$\epsilon\text{-closure}(q_2)$: From q_2 , there are no outgoing ϵ -transitions

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

Step 2: Construct the Equivalent DFA

To construct the DFA, the transition function δ_{DFA} is defined as taking the ϵ -closure of the union of the NFA transitions for a given subset of states.

1. Initial State (A):

The initial state of the DFA is the ϵ -closure of the NFA's start state.

$$A = \epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

2. Evaluate Transitions for State A ({q_0, q_1, q_2}):

On input 0:

NFA transitions from $\{q_0, q_1, q_2\}$ on 0 yield $\emptyset \cup \{q_1\} \cup \{q_2\} = \{q_1, q_2\}$.

$$\epsilon\text{-closure}(\{q_1, q_2\}) = \epsilon\text{-closure}(q_1) \cup \epsilon\text{-closure}(q_2) = \{q_1, q_2\} \cup \{q_2\} = \{q_1, q_2\}.$$

Call this new state **B**.

On input 1:

NFA transitions from $\{q_0, q_1, q_2\}$ on 1 yield $\{q_0\} \cup \emptyset \cup \{q_2\} = \{q_0, q_2\}$.

$$\epsilon\text{-closure}(\{q_0, q_2\}) = \epsilon\text{-closure}(q_0) \cup \epsilon\text{-closure}(q_2) = \{q_0, q_1, q_2\} \cup \{q_2\} = \{q_0, q_1, q_2\}.$$

This is state **A**.

3. Evaluate Transitions for State B ({q_1, q_2}):

On input 0:

NFA transitions from $\{q_1, q_2\}$ on 0 yield $\{q_1\} \cup \{q_2\} = \{q_1, q_2\}$. $\epsilon\text{-closure}(\{q_1, q_2\}) = \{q_1, q_2\}$.

This is state **B**.

On input 1:

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NFA transitions from $\{q_1, q_2\}$ on 1 yield emptyset cup $\{q_2\} = \{q_2\} \cdot \epsilon \text{ text}\{-\text{closure}\}(\{q_2\}) = \{q_2\}$.

Call this new state **C**.

4. Evaluate Transitions for State C ($\{q_2\}$):

On input 0:

NFA transitions from $\{q_2\}$ on 0 yield $\{q_2\}$.

$\epsilon \text{ text}\{-\text{closure}\}(\{q_2\}) = \{q_2\}$

This is state **C**

On input 1:

NFA transitions from $\{q_2\}$ on 1 yield $\{q_2\} \cdot \epsilon$

$\text{text}\{-\text{closure}\}(\{q_2\}) = \{q_2\}$.

This is state **C**.

Step 3: Define Final States and Transition Table

In the subset construction algorithm, any state in the DFA that contains an accepting state from the original NFA becomes an accepting state itself. Since the NFA's accepting state is q_2 , any subset containing q_2 is final.

Looking at our DFA states ($A = \{q_0, q_1, q_2\}$, $B = \{q_1, q_2\}$, $C = \{q_2\}$), **all of them contain q_2** . Therefore, all states in the resulting DFA are final/accepting states.

DFA Transition Table:

DFA State	NFA Subset Equival	Input: 0	Input: 1
$\rightarrow *$	$\{q_0, q_1, q_2\}$	B	A
* B	$\{q_1, q_2\}$	B	C
* C	$\{q_2\}$	C	C

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing(2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers some parts only	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and wellorganized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand